

PrimePy: Number Theory Library

Jonatan Beiruty

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1 Introduction

PrimePy is a C++ number theory library with Python bindings, designed for high-performance computation and symbolic experimentation.

It is structured into three primary components:

1. Analytic Number Theory
2. Algebraic Number Theory
3. Cryptographic Algorithms

The PrimePy project is guided by the following core principles:

- **Pythonic interface:** A simple and flexible Python API that feels native and encourages experimentation.
- **High-performance C++ backend:** Core computations are implemented in C++ for speed and scalability.
- **Seamless visualization:** Built-in support for intuitive and customizable visualization of number-theoretic objects and results.
- **Minimal and general abstractions:** Mathematical structures are defined in a general and reusable way, without unnecessary overhead.